

RENMARK, Australia

WMO: 946870

Lat: 34.167S

Lon: 140.750E

Elev: 69

StdP: 14.66

Time Zone: 9.50 (AUA)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.1	40.1	28.9	22.9	68.2	31.7	26.1	63.2	26.3	54.6	20.2	54.8	0.5	140	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.6	102.0	69.1	98.1	68.5	94.5	67.3	74.2	87.3	72.1	87.6	70.3	88.0	4.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	114.2	79.7	67.4	100.8	79.0	64.4	90.9	77.7	37.8	85.1	35.9	87.3	34.2	88.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.9	16.0	12.3	DB	32.9	108.9	3.3	3.0	30.6	111.1	28.7	112.8	26.8	114.5	24.5	116.7
			WB	32.8	77.5	1.9	2.7	31.4	79.4	30.3	81.0	29.2	82.5	27.8	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	76.3	77.5	70.0	62.8	57.2	53.5	52.1	53.7	59.0	64.0	68.8	73.4	(6)
	DBStd	10.58	7.33	7.31	6.93	5.51	4.63	4.09	3.50	4.41	5.93	6.67	7.45	7.00	(7)
	HDD50	42	0	0	0	0	1	11	17	12	1	0	0	0	(8)
	HDD65	1822	2	2	21	109	246	344	401	352	199	101	38	7	(9)
	CDD50	5135	814	771	621	383	226	117	81	127	272	435	563	725	(10)
	CDD65	1439	351	353	177	42	5	1	0	1	20	70	152	267	(11)
	CDH74	16958	4309	4254	1860	454	34	4	0	19	231	860	1769	3164	(12)
	CDH80	8285	2282	2290	842	127	2	0	0	2	57	343	795	1545	(13)
(14) Wind	WSAvg	3.8	4.3	3.7	3.7	2.7	2.5	2.7	3.2	4.1	4.3	4.8	4.4	4.6	(14)
(15) Precipitation	PrecAvg	10.2	0.7	0.7	0.5	0.7	0.9	0.8	1.1	1.0	1.1	1.2	0.9	0.7	(15)
	PrecMax	20.4	3.8	4.1	3.2	5.2	3.6	2.4	2.8	3.0	4.4	4.3	4.3	2.7	(16)
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	(17)
	PrecStd	3.5	0.7	0.9	0.6	0.9	0.7	0.6	0.6	0.6	0.9	1.1	0.9	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.4	105.9	100.3	88.2	78.4	71.9	70.1	76.6	86.0	96.2	100.7	102.7	(19)
		MCWB	71.7	71.9	68.1	63.1	59.4	57.0	54.3	57.1	60.2	63.4	68.0	68.0	(20)
	2%	DB	101.9	101.6	94.5	84.0	74.2	66.4	65.6	71.0	79.9	88.7	95.0	97.7	(21)
		MCWB	69.3	70.1	67.3	62.9	59.3	55.5	53.2	54.6	59.1	61.6	65.9	66.3	(22)
	5%	DB	97.2	96.9	89.5	79.8	70.3	63.4	62.5	67.1	75.1	82.7	88.9	93.4	(23)
		MCWB	69.2	69.4	66.1	62.0	58.4	55.3	52.4	53.5	58.4	60.7	63.6	65.6	(24)
	10%	DB	92.6	92.8	84.8	74.9	66.9	61.1	59.9	63.8	70.7	78.0	83.8	88.7	(25)
		MCWB	67.7	67.9	65.0	60.7	56.9	53.9	51.6	52.4	56.6	59.7	62.7	64.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	77.1	72.1	68.4	63.1	60.3	57.1	59.8	64.6	67.0	70.7	72.6	(27)
		MCDB	93.8	94.0	90.3	80.2	70.8	64.7	64.6	71.1	75.5	84.3	89.8	85.6	(28)
	2%	WB	74.1	74.9	69.7	65.5	61.5	58.0	54.9	56.9	61.8	64.3	68.4	70.3	(29)
		MCDB	88.7	87.1	87.2	76.2	69.8	64.2	61.6	67.4	74.9	80.1	87.8	85.1	(30)
	5%	WB	72.1	73.0	67.6	63.1	59.4	56.4	53.4	54.7	59.6	62.7	66.6	68.5	(31)
		MCDB	88.7	85.3	85.6	75.8	67.8	62.3	60.2	64.5	72.0	76.8	83.6	85.0	(32)
	10%	WB	70.1	70.5	65.9	61.3	57.7	54.7	52.2	53.0	57.4	60.9	64.6	66.4	(33)
		MCDB	88.0	88.7	82.9	74.2	65.6	60.1	58.9	62.2	69.7	74.8	80.6	84.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	25.2	24.6	22.7	22.3	18.4	15.4	16.6	19.1	20.5	22.8	23.2	24.6	(35)
		MCDBR	30.8	29.1	28.9	29.0	23.1	18.3	21.1	25.7	27.1	31.0	30.3	31.3	(36)
	5% WB	MCWBR	10.7	9.1	10.2	12.8	11.7	10.3	10.9	12.4	11.0	11.6	11.3	10.3	(37)
		MCWBR	23.3	21.1	24.7	25.4	18.8	16.0	18.0	22.0	24.1	24.4	25.9	23.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.342	0.338	0.321	0.313	0.300	0.293	0.292	0.303	0.334	0.334	0.344	0.335	(40)
		taud	2.498	2.533	2.567	2.563	2.583	2.590	2.589	2.543	2.445	2.462	2.467	2.507	(41)
	Ebn at Noon		316	310	305	289	276	270	277	288	295	308	313	319	(42)
		Edh at Noon	36	34	30	27	23	22	23	27	34	36	37	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2514	2250	1862	1365	961	786	849	1153	1567	2003	2288	2500	(44)
	RadStd		122	122	98	69	45	46	38	75	109	121	107	93	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+124	

Nomenclature: See separate page